

Table 1. Forecast scores and descriptive breach rates.

Forecast system	Score 95%	Score 99%	Hits 95%	Hits 99%
Common HS1250	-5.09302	-4.61463	4.67%	1.14%
Ordinary FHS1250, EUR fit	-5.16904	-4.68451	4.74%	1.00%
IN-FHS, EUR fit	-5.17835	-4.72156	4.58%	1.14%
IN-FHS reference mixture	-5.17798	-4.72930	4.25%	1.00%
Gaussian EWMA	-5.18140	-4.59084	5.17%	2.09%
Internal elliptical EWMA	-5.18215	-4.59770	5.08%	2.05%
Dynamic scalar BEKK, internal radial	-5.14745	-4.64087	3.96%	1.42%
Direct cash GARCH	-5.18043	-4.70620	5.12%	1.30%
Taylor optimized	-5.18238	-4.71927	4.83%	1.43%
Taylor–Wang relative	-5.17764	-4.70871	4.81%	1.19%
Taylor–Wang ridge	-5.19555	-4.72814	4.50%	1.36%
Equal functional mean	-5.19282	-4.73927	4.39%	1.21%
Prior-selected IN-FHS reference	-5.16555	-4.69449	4.09%	1.27%

Table 2. Reference-specific VaR and ES forecast dispersion.

Fixed cash task	VaR span 95%	ES span 95%	VaR span 99%	ES span 99%
Assets, EUR	27.32%	26.11%	26.50%	29.81%
Assets, USD	27.41%	26.34%	26.43%	30.12%
Flows, EUR	22.69%	22.54%	24.83%	26.36%
Flows, USD	22.70%	22.84%	25.51%	26.87%

Table 3. Mixture-minus-comparator score differences.

Comparator	Delta 95%	95% interval	Delta 99%	95% interval
IN-FHS, EUR fit	0.0004	[-0.0193, 0.0200]	-0.0077	[-0.0508, 0.0354]
Ordinary FHS1250, EUR fit	-0.0089	[-0.0323, 0.0145]	-0.0448	[-0.1014, 0.0119]
Gaussian EWMA	0.0034	[-0.0384, 0.0453]	-0.1385	[-0.2844, 0.0075]
Internal elliptical EWMA	0.0042	[-0.0369, 0.0452]	-0.1316	[-0.2720, 0.0088]
Taylor optimized	0.0044	[-0.0168, 0.0256]	-0.0100	[-0.0699, 0.0499]

Table 4. Paired scores against the dynamic joint control.

Tail confidence	Block length	Mixture FZ0	BEKK FZ0	Mixture minus BEKK	Simultaneous 95% interval, two-tail family
95%	20	-5.17798	-5.14745	-0.03053	[-0.05402, -0.00704]
95%	60	-5.17798	-5.14745	-0.03053	[-0.05892, -0.00214]
99%	20	-4.72930	-4.64087	-0.08843	[-0.15554, -0.02132]
99%	60	-4.72930	-4.64087	-0.08843	[-0.16342, -0.01344]

Table 5. VaR breach counts by cash target.

Forecast system	Tail	Assets EUR	Assets USD	Flows EUR	Flows USD
IN-FHS, EUR fit	95%	81	81	106	107
IN-FHS, EUR fit	99%	17	17	30	29
IN-FHS reference mixture	95%	71	72	102	103
IN-FHS reference mixture	99%	14	12	28	28
Gaussian EWMA	95%	91	90	121	121
Gaussian EWMA	99%	28	26	58	59
Internal elliptical EWMA	95%	89	90	119	118
Internal elliptical EWMA	99%	26	26	57	59
Direct cash GARCH	95%	86	87	123	123
Direct cash GARCH	99%	14	16	38	38
Dynamic scalar BEKK, internal radial	95%	72	70	91	91
Taylor optimized	95%	79	81	117	118
Dynamic scalar BEKK, internal radial	99%	19	17	40	40
Taylor optimized	99%	14	14	44	45
Taylor–Wang ridge	95%	73	74	110	111
Taylor–Wang ridge	99%	17	17	38	39

Table 6. Longer-horizon mixture breach counts.

Cash task	h=5: breaches 95% / 99%	h=20: breaches 95% / 99%
Assets, EUR	78 / 12	80 / 6
Assets, USD	78 / 12	82 / 7
Flows, EUR	101 / 32	149 / 51
Flows, USD	100 / 33	148 / 53
Financed treasury, EUR	85 / 28	107 / 40
Financed treasury, USD	85 / 26	108 / 40
Unfinanced treasury, EUR	85 / 28	107 / 40
Unfinanced treasury, USD	80 / 31	77 / 23

Table 7. Unfinanced treasury forecasts by reporting currency.

Holding intervals	Mean mixture ES95, EUR / N	Mean mixture ES95, USD / N_USD	Mean absolute realized EUR/USD normalized loss difference
1	0.006115	0.003449	0.002030
5	0.013595	0.007819	0.004559
20	0.022599	0.015020	0.008760

Table 8. Mixture breach counts in the 2026 extension.

Cash book	EUR: 95% / 99%	USD: 95% / 99%
Funded assets	2 / 1	2 / 1
Net cash flows	7 / 5	8 / 5
Unfinanced treasury	9 / 2	9 / 3

Table S1. All system/reference configurations on common dates. (continued 1/2)

Configuration	Score 95%	Score 99%	Hits 95%	Hits 99%
Common HS1250	-5.09302	-4.61463	4.67%	1.14%
Common HS500	-5.10180	-4.56166	4.78%	1.14%
Dynamic scalar BEKK, internal radial	-5.14745	-4.64087	3.96%	1.42%
Direct cash GARCH	-5.18043	-4.70620	5.12%	1.30%
Direct cash HS1250	-5.09303	-4.61436	4.67%	1.14%
Direct cash HS500	-5.10042	-4.55995	4.91%	1.34%
Equal functional mean	-5.19282	-4.73927	4.39%	1.21%
FHS1250, CHF fit	-5.14999	-4.65424	4.48%	1.00%
FHS1250, GBP fit	-5.14862	-4.70656	5.23%	1.28%
FHS1250, JPY fit	-5.17817	-4.73276	4.37%	1.09%
FHS1250, USD fit	-5.18233	-4.73908	4.66%	1.01%
FHS250, CHF fit	-5.15453	-4.60713	4.86%	1.41%
FHS250, EUR fit	-5.16503	-4.63521	4.88%	1.41%
FHS250, GBP fit	-5.15375	-4.64660	5.00%	1.34%
FHS250, JPY fit	-5.16664	-4.63530	4.92%	1.45%
FHS250, USD fit	-5.16868	-4.67036	4.83%	1.41%
FHS500, CHF fit	-5.15263	-4.63755	4.72%	1.01%
FHS500, EUR fit	-5.17484	-4.67799	4.86%	0.97%
FHS500, GBP fit	-5.15504	-4.69347	4.78%	1.08%
FHS500, JPY fit	-5.16842	-4.70863	4.73%	1.21%
FHS500, USD fit	-5.17012	-4.70683	4.56%	1.05%
Gaussian EWMA	-5.18140	-4.59084	5.17%	2.09%
IN-FHS reference mixture	-5.17798	-4.72930	4.25%	1.00%
IN-FHS, EUR fit	-5.17835	-4.72156	4.58%	1.14%
IN-FHS1250, CHF fit	-5.15050	-4.64666	4.45%	1.17%
IN-FHS1250, GBP fit	-5.14521	-4.69355	5.08%	1.31%
IN-FHS1250, JPY fit	-5.17326	-4.70638	4.17%	1.31%
IN-FHS1250, USD fit	-5.17723	-4.71413	4.26%	1.21%
IN-TCOP reference mixture	-5.17122	-4.70101	4.15%	0.98%

Table S1. All system/reference configurations on common dates. (continued 2/2)

Configuration	Score 95%	Score 99%	Hits 95%	Hits 99%
IN-TCOP, EUR fit	-5.16965	-4.68189	4.48%	1.17%
IN-TCOP1250, CHF fit	-5.13602	-4.58928	4.47%	1.34%
IN-TCOP1250, GBP fit	-5.13371	-4.62044	4.90%	1.37%
IN-TCOP1250, JPY fit	-5.17156	-4.70707	4.07%	1.09%
IN-TCOP1250, USD fit	-5.17014	-4.69793	4.12%	1.10%
Internal elliptical EWMA	-5.18215	-4.59770	5.08%	2.05%
Ordinary FHS1250, EUR fit	-5.16904	-4.68451	4.74%	1.00%
Past-best functional member	-5.16525	-4.68350	4.84%	1.22%
Prior-selected IN-FHS reference	-5.16555	-4.69449	4.09%	1.27%
Prior-selected IN-TCOP reference	-5.16005	-4.67219	4.08%	1.14%
Taylor optimized	-5.18238	-4.71927	4.83%	1.43%
Taylor–Wang relative	-5.17764	-4.70871	4.81%	1.19%
Taylor–Wang ridge	-5.19555	-4.72814	4.50%	1.36%

Table S2. Principal contrasts with 20-date circular blocks.

Tail	Comparator	Delta	Simultaneous interval
95%	IN-FHS, EUR fit	0.0004	[-0.0184, 0.0191]
95%	Ordinary FHS1250, EUR fit	-0.0089	[-0.0290, 0.0111]
95%	Gaussian EWMA	0.0034	[-0.0374, 0.0443]
95%	Internal elliptical EWMA	0.0042	[-0.0360, 0.0444]
95%	Taylor optimized	0.0044	[-0.0171, 0.0259]
99%	IN-FHS, EUR fit	-0.0077	[-0.0506, 0.0351]
99%	Ordinary FHS1250, EUR fit	-0.0448	[-0.0901, 0.0006]
99%	Gaussian EWMA	-0.1385	[-0.2806, 0.0037]
99%	Internal elliptical EWMA	-0.1316	[-0.2698, 0.0066]
99%	Taylor optimized	-0.0100	[-0.0657, 0.0456]

Table S3. Task-specific mixture-minus-comparator mean scores.

Comparator	Tail	Assets EUR	Assets USD	Flows EUR	Flows USD
IN-FHS, EUR fit	95%	0.0101	0.0098	-0.0093	-0.0091
IN-FHS, EUR fit	99%	0.0143	0.0144	-0.0307	-0.0290
Ordinary FHS1250, EUR fit	95%	0.0138	0.0142	-0.0320	-0.0318
Ordinary FHS1250, EUR fit	99%	0.0118	0.0117	-0.1025	-0.1001
Gaussian EWMA	95%	0.0242	0.0243	-0.0166	-0.0181
Gaussian EWMA	99%	-0.0271	-0.0278	-0.2467	-0.2523
Internal elliptical EWMA	95%	0.0236	0.0237	-0.0146	-0.0161
Internal elliptical EWMA	99%	-0.0279	-0.0290	-0.2321	-0.2373
Dynamic scalar BEKK, internal radial	95%	-0.01619	-0.01635	-0.04420	-0.04537
Taylor optimized	95%	0.0038	0.0047	0.0066	0.0025
Dynamic scalar BEKK, internal radial	99%	-0.01351	-0.01525	-0.16022	-0.16473
Taylor optimized	99%	0.0013	-0.0020	-0.0225	-0.0169

Table S4. Legacy EUR-normalized IN-FHS reference spreads and pointwise intervals.

Cash task	Median 95%	Pointwise interval	Median 99%	Pointwise interval
Funded assets, EUR	27.06%	[22.44%, 31.21%]	31.42%	[26.82%, 37.88%]
Funded assets, USD	27.33%	[22.60%, 31.53%]	31.94%	[27.27%, 38.46%]
Net cash flows, EUR	22.21%	[19.62%, 24.70%]	25.24%	[22.79%, 27.92%]
Net cash flows, USD	22.43%	[19.81%, 24.93%]	25.69%	[23.08%, 28.51%]

Table S5. Every annual legacy EUR-normalized IN-FHS median spread.

Cash task and tail	2018	2019	2020	2021	2022	2023	2024	2025
Funded assets, EUR, 95%	19.21	29.82	16.25	17.13	34.47	30.57	31.38	40.32
Funded assets, USD, 95%	19.36	30.30	16.37	17.27	34.62	30.81	31.57	40.35
Net cash flows, EUR, 95%	21.74	30.91	18.56	18.72	25.57	22.44	18.94	18.98
Net cash flows, USD, 95%	22.00	31.28	18.75	18.92	25.87	22.84	19.29	19.12
Funded assets, EUR, 99%	26.97	42.34	15.32	16.77	34.99	37.59	36.67	40.39
Funded assets, USD, 99%	27.98	43.44	15.54	16.92	35.25	38.22	37.33	40.73
Net cash flows, EUR, 99%	26.95	43.53	19.82	28.37	25.93	22.47	20.52	19.57
Net cash flows, USD, 99%	27.58	44.98	20.06	28.91	26.34	22.66	20.76	19.89

Table S6. Task-specific mixture calibration moments and uncertainty.

Cash task	Tail	Hit rate	Pointwise interval	Mean u	Pointwise interval
Funded assets, EUR	95%	3.47%	[2.58%, 4.36%]	0.066	[-0.016, 0.149]
Funded assets, USD	95%	3.52%	[2.59%, 4.45%]	0.068	[-0.016, 0.151]
Net cash flows, EUR	95%	4.99%	[3.67%, 6.31%]	-0.100	[-0.245, 0.045]
Net cash flows, USD	95%	5.03%	[3.72%, 6.35%]	-0.100	[-0.246, 0.046]
Funded assets, EUR	99%	0.68%	[0.32%, 1.05%]	0.029	[-0.133, 0.191]
Funded assets, USD	99%	0.59%	[0.27%, 0.90%]	0.031	[-0.133, 0.195]
Net cash flows, EUR	99%	1.37%	[0.77%, 1.96%]	-0.218	[-0.479, 0.042]
Net cash flows, USD	99%	1.37%	[0.77%, 1.96%]	-0.219	[-0.484, 0.045]

Table S7. Annual marginal parameter variation, 2018–2025.

Fitting reference	a minimum–maximum	Median a	Median b	nu minimum–maximum	Median nu
CHF	0.0037–0.1275	0.0423	0.9212	4.06–16.02	5.75
EUR	0.0151–0.1275	0.0413	0.9535	4.26–12.16	5.75
GBP	0.0037–0.1360	0.0710	0.8551	4.37–9.67	5.64
JPY	0.0045–0.1360	0.0426	0.9436	3.29–16.02	5.39
USD	0.0044–0.0987	0.0426	0.9414	3.29–12.16	5.66

Table S8. Signed reference ES contrasts in EUR reporting.

Fixed task	Tail	Fit reference versus EUR	Geometric mean ES difference	Simultaneous 95% interval
funded assets	95%	USD	+2.46%	[-2.28%, +7.43%]
funded assets	95%	GBP	+0.96%	[-3.63%, +5.76%]
funded assets	95%	JPY	+7.40%	[+2.96%, +12.04%]
funded assets	95%	CHF	+3.44%	[+0.18%, +6.81%]
net cashflows	95%	USD	-2.99%	[-6.99%, +1.19%]
net cashflows	95%	GBP	-1.02%	[-6.78%, +5.10%]
net cashflows	95%	JPY	-3.13%	[-7.61%, +1.57%]
net cashflows	95%	CHF	-0.44%	[-4.86%, +4.18%]
funded assets	99%	USD	+1.90%	[-3.10%, +7.16%]
funded assets	99%	GBP	+2.99%	[-1.68%, +7.89%]
funded assets	99%	JPY	+12.04%	[+6.87%, +17.47%]
funded assets	99%	CHF	+7.31%	[+2.05%, +12.84%]
net cashflows	99%	USD	-8.92%	[-12.21%, -5.50%]
net cashflows	99%	GBP	-1.34%	[-7.61%, +5.36%]
net cashflows	99%	JPY	-11.75%	[-15.17%, -8.20%]
net cashflows	99%	CHF	+0.76%	[-4.89%, +6.75%]

Table S9. Descriptive within-date log-ES partition.

Book	Tail	Reference share	Reporting-currency share	Interaction share
funded assets	95%	99.871%	0.125%	0.005%
funded assets	99%	99.792%	0.196%	0.012%
net cashflows	95%	99.840%	0.152%	0.007%
net cashflows	99%	99.756%	0.227%	0.017%

Table S10. Exploratory numerical margin requirements.

Tail	Comparator	Smallest numerical NI margin (block 60)	Smallest numerical two-direction margin (block 60)
95%	IN_FHS1250@EUR	0.01683	0.01706
95%	FHS1250@EUR	0.01065	0.02966
95%	EWMA_GAUSS	0.03847	0.03847
95%	EWMA_SELFNORMALIZED	0.03856	0.03856
95%	TAYLOR_OPT	0.02217	0.02217
99%	IN_FHS1250@EUR	0.02806	0.04762
99%	FHS1250@EUR	0.00228	0.09723
99%	EWMA_GAUSS	0.00000	0.27358
99%	EWMA_SELFNORMALIZED	0.00000	0.26162
99%	TAYLOR_OPT	0.03973	0.06549

Table S11. Conditional parameter perturbations of symmetric R_ES.

Fixed origin	EUR-reported funded assets tail	Block	Original R_ES	Parameter-perturbation median	2.5–97.5 percentiles
2018-01-02	95%	20	17.74%	17.96%	[6.70%, 32.13%]
2018-01-02	99%	20	29.84%	17.28%	[6.95%, 44.23%]
2021-01-04	95%	20	39.25%	18.27%	[7.04%, 32.92%]
2021-01-04	99%	20	37.00%	18.52%	[7.58%, 33.38%]
2025-01-02	95%	20	16.97%	16.82%	[8.65%, 26.36%]
2025-01-02	99%	20	26.06%	23.12%	[15.98%, 31.34%]
2018-01-02	95%	60	17.74%	18.49%	[5.40%, 35.42%]
2018-01-02	99%	60	29.84%	17.43%	[6.78%, 47.11%]
2021-01-04	95%	60	39.25%	27.13%	[13.13%, 46.47%]
2021-01-04	99%	60	37.00%	26.84%	[13.23%, 44.08%]
2025-01-02	95%	60	16.97%	16.58%	[9.23%, 25.48%]
2025-01-02	99%	60	26.06%	25.16%	[18.53%, 32.24%]

Table S12. Symmetric reference diagnostics in every extended cash task.

Evaluation	h	Cash task	Median R_ES 95%	Median R_ES 99%
temporal_2026	1	Assets, EUR	21.99%	22.08%
temporal_2026	1	Assets, USD	21.91%	22.18%
temporal_2026	1	Flows, EUR	22.75%	27.13%
temporal_2026	1	Flows, USD	23.16%	27.35%
horizon_historical	5	Financed treasury, EUR	21.64%	32.76%
horizon_historical	5	Financed treasury, USD	22.23%	34.08%
horizon_historical	5	Assets, EUR	26.40%	31.85%
horizon_historical	5	Assets, USD	26.86%	32.93%
horizon_historical	5	Flows, EUR	23.02%	24.71%
horizon_historical	5	Flows, USD	23.62%	25.61%
horizon_historical	20	Financed treasury, EUR	22.02%	28.13%
horizon_historical	20	Financed treasury, USD	22.68%	29.33%
horizon_historical	20	Assets, EUR	24.09%	28.10%
horizon_historical	20	Assets, USD	25.20%	28.72%
horizon_historical	20	Flows, EUR	24.85%	27.92%
horizon_historical	20	Flows, USD	25.22%	28.76%
unfinanced_historical	1	Unfinanced treasury, EUR	19.63%	25.23%
unfinanced_historical	1	Unfinanced treasury, USD	15.33%	26.00%
unfinanced_historical	5	Unfinanced treasury, EUR	21.64%	32.76%
unfinanced_historical	5	Unfinanced treasury, USD	13.90%	20.78%
unfinanced_historical	20	Unfinanced treasury, EUR	22.02%	28.13%
unfinanced_historical	20	Unfinanced treasury, USD	14.75%	19.99%
unfinanced_2026	1	Unfinanced treasury, EUR	27.36%	27.17%
unfinanced_2026	1	Unfinanced treasury, USD	20.33%	23.78%

Table S13. Historical MCS families and both block specifications.

Candidate family	Tail	Retained, blocks 20 / 60	Mixture inclusion p, 20 / 60
all_41	95%	33 / 33	0.381 / 0.359
exclude_standalone_IN_TCOP	95%	28 / 28	0.363 / 0.359
all_41	99%	36 / 36	0.877 / 0.862
exclude_standalone_IN_TCOP	99%	29 / 28	0.877 / 0.862
42: all plus BEKK	95%	33 / 33	0.381 / 0.359
35: exclude standalone IN-TCOP	95%	28 / 28	0.363 / 0.359
42: all plus BEKK	99%	36 / 37	0.877 / 0.862
35: exclude standalone IN-TCOP	99%	29 / 30	0.877 / 0.862

Table S14. Temporal and longer-horizon MCS specifications.

Evaluation	Horizon	Tail	Date block	Dates	Candidate count	Retained	Mixture MCS p-value
temporal	1	95%	20	189	9	7	0.3131
temporal	1	95%	60	189	9	5	0.0328
temporal	1	99%	20	189	9	9	0.1056
temporal	1	99%	60	189	9	2	0.0100
horizons	5	95%	20	2042	7	7	1.0000
horizons	5	95%	60	2042	7	7	1.0000
horizons	5	99%	20	2042	7	7	1.0000
horizons	5	99%	60	2042	7	7	1.0000
horizons	20	95%	20	2027	7	7	0.7341
horizons	20	95%	60	2027	7	7	0.7858
horizons	20	95%	120	2027	7	7	0.7504
horizons	20	99%	20	2027	7	7	0.9542
horizons	20	99%	60	2027	7	7	0.9564
horizons	20	99%	120	2027	7	7	0.9536

Table S15. Symmetric diagnostic precision and upper quantiles.

Cash task	Tail	Median R_ES	Pointwise 95% median interval	90th / 99th percentiles
funded assets, EUR	95%	26.11%	[21.67%, 30.20%]	47.55% / 67.49%
funded assets, EUR	99%	29.81%	[25.40%, 35.66%]	49.89% / 71.05%
funded assets, USD	95%	26.34%	[21.91%, 30.44%]	47.91% / 68.20%
funded assets, USD	99%	30.12%	[25.79%, 36.16%]	50.80% / 72.15%
net cashflows, EUR	95%	22.54%	[19.74%, 25.44%]	37.64% / 59.60%
net cashflows, EUR	99%	26.36%	[24.27%, 29.11%]	40.80% / 57.53%
net cashflows, USD	95%	22.84%	[19.93%, 25.66%]	38.01% / 61.14%
net cashflows, USD	99%	26.87%	[24.55%, 29.77%]	41.90% / 58.78%

Table S16. Absolute and symmetric ES spans.

Fixed cash task	R_ES 95%	D_ES 95%	R_ES 99%	D_ES 99%
Assets, EUR	26.11%	0.159%	29.81%	0.271%
Assets, USD	26.34%	0.162%	30.12%	0.279%
Flows, EUR	22.54%	0.101%	26.36%	0.177%
Flows, USD	22.84%	0.103%	26.87%	0.182%

Table S17. Companion native VaR and ES dispersion.

Fixed cash task	VaR span 95%	ES span 95%	VaR span 99%	ES span 99%
Assets, EUR	27.32%	26.11%	26.50%	29.81%
Assets, USD	27.41%	26.34%	26.43%	30.12%
Flows, EUR	22.69%	22.54%	24.83%	26.36%
Flows, USD	22.70%	22.84%	25.51%	26.87%